Story 4: Data Sources, Methods & Visual Choices

DATA 608 | Muhammad Suffyan Khan | October 2026 • Accompanies the final 10-slide presentation

Research question & analytical purpose

How does annual mean wage vary across data-related occupational categories and U.S. states? This is a descriptive comparison of published occupational wage estimates, not a prediction of individual pay and not a causal study.

Data source & provenance

The statistical origin is the U.S. Bureau of Labor Statistics (BLS), May 2025 Occupational Employment and Wage Statistics (OEWS). The numerical observations included here were transcribed from SignalBleed occupation-by-state pages labeled as BLS OEWS May 2025, because the official BLS bulk files were not obtained during preparation. They are third-party BLS-attributed transcriptions, not direct official BLS exports. Each curated record includes a source URL; the original official tables are listed under Sources below.

Occupation definitions and geographic coverage

The six detailed Standard Occupational Classification (SOC) groups are: Data Scientists (15-2051), Database Architects (15-1243), Database Administrators (15-1242), Computer Systems Analysts (15-1211), Operations Research Analysts (15-2031), and Statisticians (15-2041). Common industry titles including “Data Engineer,” “Data Analyst,” and “Business Analyst” are not exact one-to-one equivalents to these SOC groups and are not presented as such.

The joint occupation-by-state dataset contains 16 deliberately selected states × six occupations (96 rows), including one unavailable wage entry (New Jersey / Operations Research Analysts). A separate Data Scientist table contains 49 published jurisdictions from its source. Comparisons across the 16-state subset are not representative national estimates.

Measure, calculations & cautions

The unit is nominal annual mean employee wage in U.S. dollars. The occupation ranking is an arithmetic average of available state mean wages across the 16 selected states, weighting states equally. It is not an employment-weighted U.S. mean. Missing observations are retained as missing (NaN), never replaced with zero. The within-state gap is Data Scientist mean wage minus Computer Systems Analyst mean wage. The fifth visualization reports each occupation’s lowest and highest published wages within the same 16-state selection and the median of those state-level wage estimates.

Wages do not incorporate cost-of-living, taxes, worker experience, individual job titles, bonuses, or benefits. Geographic differences do not establish causes or differences in purchasing power.

Story development & audience

The intended audience is classmates and readers interested in data-related careers. The sequence moves from role ranking, to joint state/role patterns, to the geographic extremes of one occupation, to within-state wage gaps, to the range of geographic dispersion by role; a dedicated conclusion then synthesizes distinct findings. Each chart is paired with a direct takeaway to reduce interpretation effort.

Visualization 1: Ranked occupation comparison (Slide 3)

Sorted horizontal bars make six occupations comparable on a common dollar scale. Database Architects are emphasized in orange, with other occupations in muted blue, to draw attention to the highest equal-state average (about $142.5k). Direct labels remove dependence on a legend. This chart establishes the role dimension; it does not claim to show national means.

Visualization 2: Occupation × state heatmap (Slide 4)

A labeled matrix encodes annual mean wage for each occupation within each of 16 states. A sequential teal-to-navy scale represents magnitude consistently rather than implying good/bad categories. The single unpublished cell is rendered gray or marked unavailable, not zero. Reading across rows reveals occupation differences within a state; reading down columns shows geographic differences for a role.

Visualization 3: Geographic extremes for Data Scientists (Slide 5)

A ranked horizontal bar chart shows the five highest and five lowest published jurisdiction mean wages for Data Scientists. Selective orange emphasis isolates Washington ($158,040) and Mississippi ($79,290), a $78,750 gap and approximately 1.99× ratio. This is an intentionally focused view of the tails of the 49-jurisdiction source, not the complete geographic distribution on one slide.

Visualization 4: Within-state occupational pay gap (Slide 6)

A diverging bar chart subtracts Computer Systems Analyst wages from Data Scientist wages in the 16 states. The zero baseline is substantively important: positive values favor Data Scientists; negative values favor Systems Analysts. Teal and orange separate the signs, revealing three states with negative differences. This tests a distinct question—whether the relative advantage persists everywhere.

Visualization 5: Wage dispersion by occupation (Slide 7)

A range plot shows each occupation's minimum and maximum selected-state mean wage, connected by a horizontal interval; an orange marker indicates the median selected-state mean. Unlike the simple role ranking, it reveals how much geographic spread exists within a role. All endpoints and medians refer to the same 16 selected states, not the national wage distribution or an individual-worker range.

Salience, simplicity & presentation uniformity

The final slides use a consistent navy/teal/orange visual system, readable typography, meaningful titles, direct numerical labels, and restrained gridlines. Emphasis is tied to findings rather than decoration. Text interpretation panels explain both what to notice and what not to infer. Separate charts communicate complementary messages instead of repeating the same claim.

Conclusion & instructor feedback integration

Previous instructor feedback praised messaging and selective highlighting, while requesting stronger attention guidance, varied chart forms, distinct messages, and an explicit conclusion. The final presentation now contains five analytical visualizations with separate purposes and a clearly labeled Conclusion on Slide 9. That slide synthesizes occupational ranking, Data Scientist geography, and state-dependent occupational wage gaps, improving the ease with which a viewer can understand the overall story (Amity).

Reproducibility & limitations

The companion Python notebook and local processed CSVs support the calculations and visualizations; a separate generation script is retained in the project. The slide PDF is exported from the same PowerPoint. Data values were compiled from published third-party BLS-attributed tables rather than freshly retrieved through a BLS API. The geographic selection and missing-data limitations should accompany any reuse of the findings.

Sources

https://www.bls.gov/oes/tables.htm
https://www.bls.gov/oes/oes_doc.htm
https://signalbleed.com/jobs/data-scientists.html
https://signalbleed.com/jobs/database-architects.html
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